

# BIOLOGICAL ANTHROPOLOGY - LABORATORY ACTIVITIES (TAUB)



Book: Biological Anthropology - Laboratory  
Activities (Taub)

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### Contents:

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**Please note:** These activities are meant to be used in connection with a full Biological Anthropology Course and full textbook. The instructor is expected to present the base material that students will need to complete each activity. This allows the instructor to mold the activities to their own approach. Students will need an assigned text to assist with these activities, identify bone and features, understand the proper use of Hardy-Weinberg Equilibrium, significance of primate taxonomy, and specific information about various early human forms. These activities are presented in the sequence they might be used if teaching with Introduction to Physical Anthropology by Robert Jurmain et al.<sup>1</sup> These activities do not need to be used in the order presented and are meant to allow an instructor to have lab activities for students that do not require the students to purchase a separate lab manual. The contents of this manuscript are available by means of Creative Commons. They may be used free of charge as long as this author and all other copyright holders are given credit for their work. Should you have any activities you wish to add to this manual, and make available under creative commons to other instructors, please notify this author.

### [Lab 1 & 2](#) Pages 5-12

These are the only lab quizzes in this book. On the first week I introduce the students to the 206 bones of the typical adult human. Week two, the students are provided with blank copies of the images in these activities. I then point to ten or twenty specific bones and the student must locate and name the bone on their sheets. After the quiz, I then introduce them to the major features and landmarks of the bones. Week 3, the students receive the same blank versions of these images, and again I point to 10 to 20 different features or landmarks on various bones. They are expected to locate these on the correct image, location, and name them correctly. We then continue onto lab 3.

### [Lab 3](#) Pages 13-14

This is an introduction into the application of the Hardy-Weinberg Equilibrium. It is expected that the presenter will explain the assumptions required for Hardy Weinberg equilibrium to function. The first four questions require the students to place the assigned percentages into the provided equations. The second page asks to interpret what might happen given certain situations. The goal with these activities is for students to understand that just because a trait is not “dominant” does not mean it will quickly be removed from the gene pool of a population. Sometimes, recessive traits have a better possibility for long term survival. I usually collect this lab at the start of week 4.

### [Lab 4](#) Pages 15-16

This activity takes last week’s introduction of the Hardy-Weinberg Equilibrium and applies it to monogenetic traits within your class’s population. Students are expected to solve the equations to find the expected population with a heterozygous genotype for the trait in question. They are then asked to compare their results to that of the population at large through internet research. The web address for: OMIM ® - Online Mendelian Inheritance in Man, ® is provided at the end of the activity. I collect this assignment at the beginning of week 5

### [Lab 5 & 6](#) Pages 17-24

Primate geography, social systems and diet are the focus of these activities. The class is broken into groups of 2-4 students. Each group selects on primate based on the images provided. At week 6 each group presents to the class information about the location, social behaviors, diet and physical characteristics of the primate they selected. I grade each group on their research, effort and presentation. After the presentations, each group is expected to complete the rest of the worksheet for the primates that were not presented by other groups. The students are provided with two web links that can assist them in collecting the required information. The final group worksheet is due at the beginning of week 7.

### [Lab 7](#) Pages 25-28

Students will focus on modern Human bone interpretation. In my classroom we have a real skeleton that was collected prior to 1960. The students are asked to study this skeleton and find the sex, age at death, pre-mortem injuries and post-mortem damage.

Students are asked to examine a collection of fetal skull casts and describe the major changes in skull development prior to birth. Finally, students are expected to go to the library and do internet/library research on two recent cases of historic bone analysis.

#### [Lab 8, 9, & 10 Pages 29-38](#)

The class will again split into groups of 2-4 students. Each group will start to examine a cast of various fossil or modern human skulls. In week 8, I provide casts of *Saellanthropus tchadensis*, *Artipithicus ramidus* (Ardi), *Australopithecus africanus* (Mrs. Plies) and *Australopithecus afarensis* (Lucy or similar). The students are provided with calipers, rulers and magnifying lenses. In week 9, the students receive casts of *Homo habilis*, *Homo erectus*, *Archaic Homo sapiens* and *Homo sapiens neanderthalensis* (my bias). In week 10 I provide the class with 4 different modern human casts. Students are then asked to compare the differences from week 10 with the earlier two activities.

#### [Chart: Possible Early Humans Ancestors...or not Page 39](#)

#### [Works Cited for chart Page 40](#)

## Licensing

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## CHAPTER OVERVIEW

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## 1.1: Dedication and Thanks

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### Dedication and Thanks:

First I would like to thank Dr. John Asoszatia-Petheo, Ph.D. my first Biological Anthropology instructor who had the courage to teach us and take risks. I wish to thank Professor Barbara Oldham for her assistance collecting the creative commons and copyright free images needed to make this work possible. Thanks should also go to the students (test subjects) who have helped me refine these assignments over the past few years. This work would not be available to others without the technical assistance and persistence of Dr. Claver Hategekimana, Ph.D. Many people contributed their work and photography free of charge or with a creative commons usage. Frans de Waal and Anne Zeller both agreed to allow their copyrighted primate images to be used in this effort. Others permitted their images to be used under creative commons, and are credited next to their work and I wish to thank them for their generosity. I need to thank Dr. Karl Polivka, Ph.D, for assisting with the proofreading. Finally, all thanks should go to Amanda H. S. Taub, my wife and domestic financial sponsor for allowing me to research, teach and do my various creative follies without demanding that I find a “real job.”

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## 1.2: Identifying Bones I

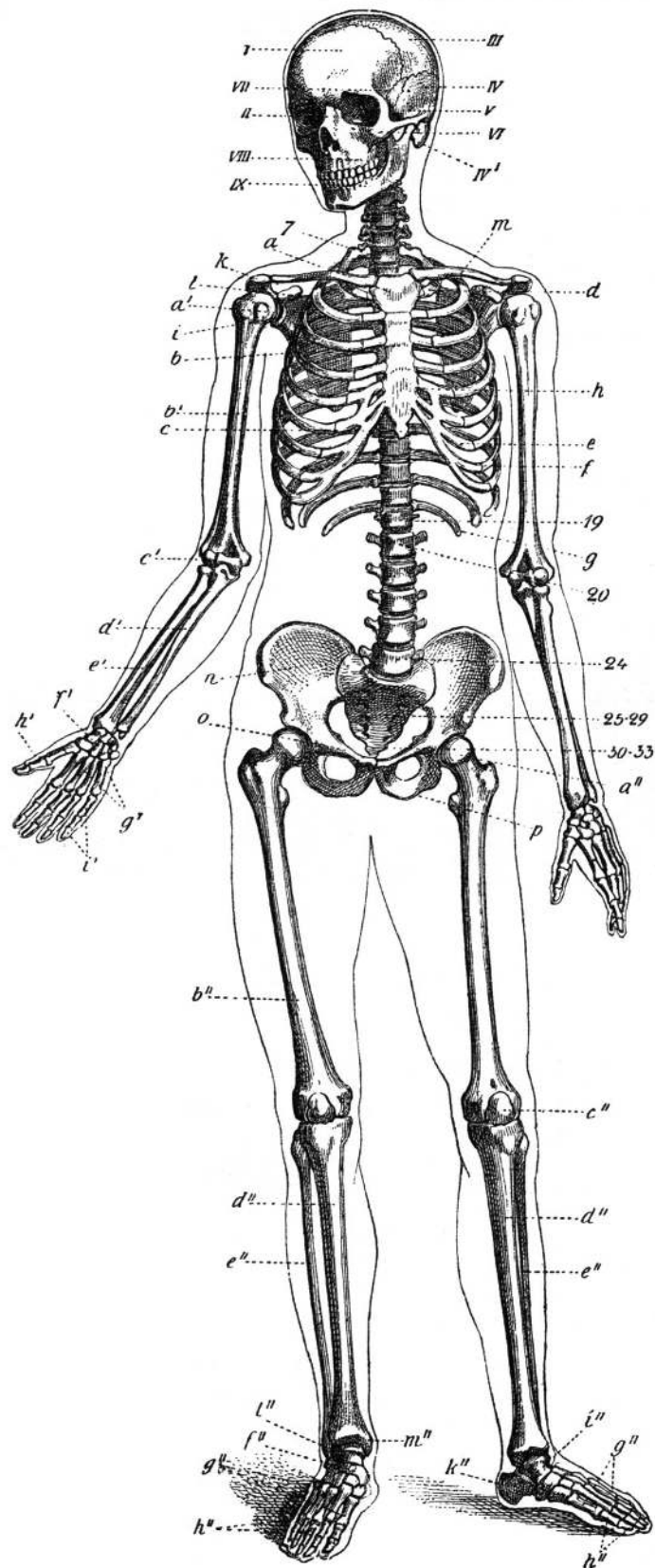
---

Name: \_\_\_\_\_

Section: \_\_\_\_\_

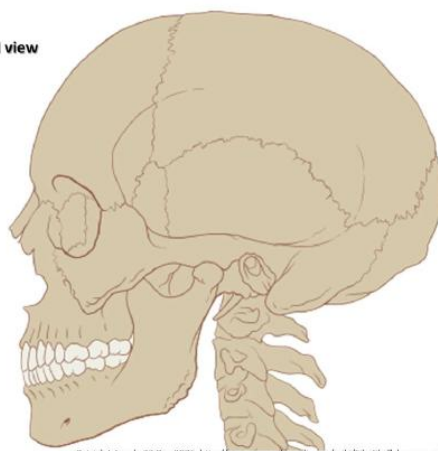
Student ID#: \_\_\_\_\_

Your instructor will provide you with the names of all 206 bones of the modern human body. You will need to remember these and be prepared for a quiz on these bones, their location and function by the next time your lab group meets.



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**Skull.**  
Left lateral view



14

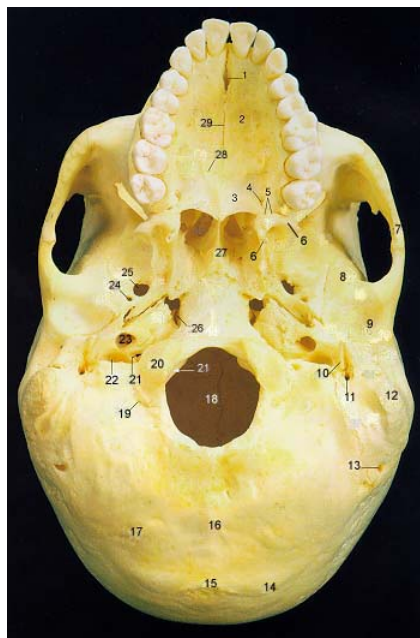
Patrick J. Lynch, 23 Dec 2006, [http://commons.wikimedia.org/wiki/File:Skull\\_human\\_and\\_cervical\\_spine.svg](http://commons.wikimedia.org/wiki/File:Skull_human_and_cervical_spine.svg)



**Skull.**  
Anterior view

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